SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

*I **Sri Janani B (192421036)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Sri Janani B

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

ANSWER:

Given Network: 192.168.100.0/24

School of Computing (/25)

- **Subnet Mask:** 255.255.255.128
- **Network Address:** 192.168.100.0
- **Broadcast Address:** 192.168.100.127
- **Valid Host Range:** 192.168.100.1 – 192.168.100.126
- **Usable Hosts:** $2^{(32-25)} - 2 = 2^7 - 2 = 126$

School of Electronics (/26)

- **Subnet Mask:** 255.255.255.192
- **Network Address:** 192.168.100.128
- **Broadcast Address:** 192.168.100.191
- **Valid Host Range:** 192.168.100.129 – 192.168.100.190
- **Usable Hosts:** $2^{(32-26)} - 2 = 2^6 - 2 = 62$

School of Mechanical Engineering (/27)

- **Subnet Mask:** 255.255.255.224
- **Network Address:** 192.168.100.192
- **Broadcast Address:** 192.168.100.223
- **Valid Host Range:** 192.168.100.193 – 192.168.100.222
- **Usable Hosts:** $2^{(32-27)} - 2 = 2^5 - 2 = 30$

Remaining Unused IP Address Range

After allocating the three subnets, the remaining unused block is:

- **Network Address:** 192.168.100.224
- **Host Range:** 192.168.100.225 – 192.168.100.254
- **Broadcast Address:** 192.168.100.255

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**
Quality Assurance → **/25**
Human Resources → **/26**
Executive Management → **/27**
As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

ANSWER:

Given Network: 10.10.0.0/16

Software Development (/24)

- **Subnet Mask:** 255.255.255.0
- **Network Address:** 10.10.0.0
- **Broadcast Address:** 10.10.0.255
- **Host Range:** 10.10.0.1 – 10.10.0.254
- **Usable Hosts:** $2^{(32-24)} - 2 = 254$

Quality Assurance (/25)

- **Subnet Mask:** 255.255.255.128
- **Network Address:** 10.10.1.0
- **Broadcast Address:** 10.10.1.127
- **Host Range:** 10.10.1.1 – 10.10.1.126
- **Usable Hosts:** $2^{(32-25)} - 2 = 126$

Human Resources (/26)

- **Subnet Mask:** 255.255.255.192
- **Network Address:** 10.10.1.128
- **Broadcast Address:** 10.10.1.191
- **Host Range:** 10.10.1.129 – 10.10.1.190

- **Usable Hosts:** $2^{(32-26)} - 2 = 62$

Executive Management (/27)

- **Subnet Mask:** 255.255.255.224
- **Network Address:** 10.10.1.192
- **Broadcast Address:** 10.10.1.223
- **Host Range:** 10.10.1.193 – 10.10.1.222
- **Usable Hosts:** $2^{(32-27)} - 2 = 30$

Remaining Unused IP Address Range

- **Unused Range:** 10.10.1.224 – 10.10.255.255

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:
2001:0db8:abcd:0000:0000:0000:0000:0000/64
As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

ANSWER:

Given IPv6 Address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed IPv6 Address

2001:db8:abcd::/64

2. Expanded IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. Network Prefix

The network prefix is:

2001:db8:abcd:0000::/64

The first **64 bits** represent the network portion.

4. Number of Host Bits

Host bits = **128 − 64 = 64 bits**

5. Total Host Addresses

Total possible host addresses:

2^{64} = 18,446,744,073,709,551,616

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4.The **default route** that should be used if the destination does not match any routing table entry.

ANSWER:

Given Routing Table

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP: 192.168.2.45

1. Destination Subnet

The destination IP belongs to the subnet **192.168.2.0/24**.

2. Matching Routing Table Entry

The matching routing table entry is **192.168.2.0/24**.

3. Next-Hop Network/Interface

The packet will be forwarded through the interface connected to the **192.168.2.0/24** network.

4. Default Route

If no route matches, the router forwards the packet using the default route:

0.0.0.0

5.A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.

3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

ANSWER:

Administration LAN (192.168.10.0/26)

- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.63
- **Valid Host Range:** 192.168.10.1 – 192.168.10.62
- **Suggested PC Addresses:**
  - PC1: 192.168.10.2
  - PC2: 192.168.10.3
  - PC3: 192.168.10.4
- **Default Gateway:** 192.168.10.1

Finance LAN (192.168.10.64/26)

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127
- **Valid Host Range:** 192.168.10.65 – 192.168.10.126
- **Suggested PC Addresses:**
  - PC1: 192.168.10.66
  - PC2: 192.168.10.67
  - PC3: 192.168.10.68
- **Default Gateway:** 192.168.10.65

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand